TabuLLM: AI Benchmark Report

1. Executive Summary: Consolidated Audit Results

This report consolidates the results of the TabuLLM data augmentation pipeline. The objective is to audit the fidelity, diversity, and privacy of the synthetic data generated via LLM inference. Below are the averaged performance indicators across all processed datasets in this session.

Audit Indicator	Global Session Mean (± Std)
Synthetic Fidelity (JSD)	82.6 ± 1.11
Correlation Stability	88.3 ± 0.46
Distribution Consistency	88.6 ± 0.80
Entropy Preservation	82.1 ± 0.54
Mean Z Score Drift	0.3 ± 0.02
Categorical Fidelity	0.0 ± 0.00
Privacy Risk	19.2 ± 2.44
Data Density	100.0 ± 0.00
Coverage Score	40.6 ± 2.15
Feature Accuracy	70.1 ± 1.37
Novelty Score	0.5 ± 0.02

TabuLLM: AI Benchmark Report

Dataset Audit: indian_diabetes.csv

1. Dataset Portrait

Attribute	Value
Rows	768
Columns	9
Numeric features	9
Categorical features	0
Missing cells (%)	0.0%

2. Generation Strategy & LLM Config [1]

Parameter	Configuration
Algorithm (*12)	gpt-4o-mini
Folds (*13)	10
Context % (*14)	0%
Generation % (*15)	100%
Join context (*16)	False

3. Statistical Significance & Data Drift

Feature	Orig Mean	Aug Mean	Shift	P-Value	Status
Pregnancies	3.845	4.213	+0.368	9.6%	OK
Glucose	120.895	126.396	+5.502	4.6%	OK
BloodPressure	69.105	71.112	+2.007	2.9%	OK
SkinThickness	20.536	23.213	+2.677	13.0%	Warn
Insulin	79.799	91.018	+11.218	14.1%	Warn
BMI	31.993	33.967	+1.975	6.2%	OK
DiabetesPedigreeFunction	0.472	0.563	+0.091	19.4%	Warn
Age	33.241	35.071	+1.830	5.5%	OK
Outcome	0.349	1.000	+0.651	186.6%	Drift

4. Quality & Fidelity Metrics (Mean ± Std)

Metric Category	Aggregated Score
Synthetic Fidelity (JSD) (*1)	82.6 ± 1.11
Correlation Stability (*2)	88.3 ± 0.46
Distribution Consistency (*3)	88.6 ± 0.80
Entropy Preservation (*4)	82.1 ± 0.54
Mean Z Score Drift (*5)	0.3 ± 0.02
Categorical Fidelity (*6)	0.0 ± 0.00
Privacy Risk (*7)	19.2 ± 2.44
Data Density (*8)	100.0 ± 0.00
Coverage Score (*9)	40.6 ± 2.15
Feature Accuracy (*10)	70.1 ± 1.37

Novelty Score (*11)	0.5 ± 0.02

[1] (*12) Algorithm: Contextual constraint or model parameter.

[2] (*13) Folds: Contextual constraint or model parameter.

[3] (*14) Context %: Contextual constraint or model parameter.

[4] (*15) Generation %: Contextual constraint or model parameter.

[5] (*16) Join context: Contextual constraint or model parameter.

[6] Synthetic Fidelity (JSD): Overall distribution similarity.
[7] Correlation Stability: Multivariate relationships preserved.
[8] Distribution Consistency: Median feature similarity.
[9] Entropy Preservation: Information diversity maintained.
[10] Mean Z Score Drift: Standardized distribution shift.
[11] Categorical Fidelity: Category distribution similarity.
[12] Privacy Risk: Synthetic rows too close to originals.
[13] Data Density: Non-missing cells in augmented data.
[14] Coverage Score: Synthetic coverage of original space.
[15] Feature Accuracy: Mean feature similarity.
[16] Novelty Score: Distance ratio to nearest neighbors.

6. Downstream ML Utility: Comparative Analysis

Model: GB

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.7467532467532467	N/A	0.7350649350649351	-0.011688311688311682	
F1	0.6156057130748699	N/A	0.6472420844724223	0.0316363713975523	

Model: XGB

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.7292207792207792	N/A	0.7233766233766235	-0.005844155844155851	
F1	0.592152063889281	N/A	0.626151481865094	0.03399941797581294	*

Model: DT

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.7090909090909091	N/A	0.6733766233766234	-0.0357142857142857	*
F1	0.5712908396419302	N/A	0.5707313285189446	-0.00055951112298565 64	

Model: NB

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.7636363636363636	N/A	0.7448051948051948	-0.01883116883116883	*
F1	0.6399852354635025	N/A	0.6495272096073774	0.009541974143874954	

Model: SVM

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.7564935064935064	N/A	0.7305194805194806	-0.025974025974025972	
F1	0.5733680734057942	N/A	0.6483780697158745	0.07500999631008026	*

Model: AB

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.

ACCURACY	0.7493506493506492	N/A	0.7461038961038962	-0.00324675324675324	
F1	0.6076604413342895	N/A	0.669052184892337	0.06139174355804735	*

Model: RF

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.7551948051948052	N/A	0.7512987012987014	-0.00389610389610390 87	
F1	0.6160764484610317	N/A	0.6704009457820723	0.05432449732104068	*

Model: LR

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.772077922077922	N/A	0.7233766233766233	-0.04870129870129871	*
F1	0.6262914445612621	N/A	0.6447874649467294	0.01849602038546731 6	

Model: MLP

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.6558441558441558	N/A	0.6616883116883118	0.00584415584415584 1	
F1	0.5139718192589408	N/A	0.5768658418327486	0.06289402257380773	*

Model: KNN

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.7545454545454545	N/A	0.6772727272727274	-0.0772727272727273	*
F1	0.5998259463389416	N/A	0.6245223127260473	0.02469636638710571 6	

Appendix: Benchmark Data Trace

Legend: [Blue] = Real Context | [Green] = Generated Synthetic | [Yellow] = Real Training (Remaining) | [Grey] = Real Test Baseline

A. Baseline: Original Dataset (Full Record Sample)

#	Pregnancies	Glucose	BloodPressure	SkinThickness	Insulin	BMI	DiabetesPedigreeFunct	Age	Outcome
1	6.0	148.0	72.0	35.0	0.0	33.6	0.627	50.0	1.0
2	1.0	85.0	66.0	29.0	0.0	26.6	0.351	31.0	0.0
3	8.0	183.0	64.0	0.0	0.0	23.3	0.672	32.0	1.0
4	1.0	89.0	66.0	23.0	94.0	28.1	0.167	21.0	0.0
5	0.0	137.0	40.0	35.0	168.0	43.1	2.288	33.0	1.0
6	5.0	116.0	74.0	0.0	0.0	25.6	0.201	30.0	0.0
7	3.0	78.0	50.0	32.0	88.0	31.0	0.248	26.0	1.0
8	10.0	115.0	0.0	0.0	0.0	35.3	0.134	29.0	0.0
9	2.0	197.0	70.0	45.0	543.0	30.5	0.158	53.0	1.0
10	8.0	125.0	96.0	0.0	0.0	0.0	0.232	54.0	1.0
11	4.0	110.0	92.0	0.0	0.0	37.6	0.191	30.0	0.0
12	10.0	168.0	74.0	0.0	0.0	38.0	0.537	34.0	1.0
13	10.0	139.0	80.0	0.0	0.0	27.1	1.441	57.0	0.0
14	1.0	189.0	60.0	23.0	846.0	30.1	0.398	59.0	1.0
15	5.0	166.0	72.0	19.0	175.0	25.8	0.587	51.0	1.0
16	7.0	100.0	0.0	0.0	0.0	30.0	0.484	32.0	1.0
17	0.0	118.0	84.0	47.0	230.0	45.8	0.551	31.0	1.0
18	7.0	107.0	74.0	0.0	0.0	29.6	0.254	31.0	1.0
19	1.0	103.0	30.0	38.0	83.0	43.3	0.183	33.0	0.0
20	1.0	115.0	70.0	30.0	96.0	34.6	0.529	32.0	1.0

TabuLLM: AI Benchmark Report

B.1 Audit Trace: Fold 1 | Gen: 100%

#	Pregnancies	Glucose	BloodPressure	SkinThickness	Insulin	BMI	DiabetesPedigreeFunc	Age	Outcome	_fold_id
1	7.0	149.0	83.0	30.0	156.0	33.3292297139996 3	0.89129711229322 02	35.0	1.0	1.0
2	3.0	124.0	74.0	0.0	0.0	35.0181144388253 86	0.18775267124763 134	53.0	1.0	1.0
3	5.0	118.0	0.0	0.0	0.0	31.6695289129974 64	0.37922266821877 976	26.0	1.0	1.0
4	8.0	106.0	0.0	0.0	0.0	17.4308481632875 96	0.39056930468043 78	31.0	1.0	1.0
5	8.0	104.0	77.0	34.0	0.0	34.1648794996570 3	0.85979144294718 29	44.0	1.0	1.0
6	0.0	145.0	70.0	0.0	0.0	37.6992782335227 1	0.33863890304747 51	27.0	1.0	1.0
7	8.0	110.0	76.0	0.0	0.0	27.8	0.237	58.0	0.0	nan
8	1.0	90.0	62.0	18.0	59.0	25.1	1.268	25.0	0.0	nan
9	2.0	56.0	56.0	28.0	45.0	24.2	0.332	22.0	0.0	nan
10	5.0	111.0	72.0	28.0	0.0	23.9	0.407	27.0	0.0	nan

B.2 Audit Trace: Fold 2 | Gen: 100%

#	Pregnancies	Glucose	BloodPressure	SkinThickness	Insulin	BMI	DiabetesPedigreeFunc	Age	Outcome	_fold_id
1	2.0	170.0	72.0	34.0	253.0	34.5547950347974 4	0.74468830235625 69	28.0	1.0	2.0
2	8.0	52.0	77.0	35.0	0.0	36.6238224600509 24	0.78379643875975 84	49.0	1.0	2.0
3	8.0	32.0	69.0	37.0	0.0	35.3773398231418 8	0.55950065848935 58	42.0	1.0	2.0
4	2.0	159.0	71.0	23.0	125.0	29.6241923340724 3	0.30523403490022 194	36.0	1.0	2.0
5	4.0	144.0	67.0	0.0	0.0	36.0538429450950 3	0.20784593458149 359	36.0	1.0	2.0
6	7.0	151.0	78.0	33.0	210.0	42.4488509100489 6	0.50040313146169 28	35.0	1.0	2.0
7	0.0	91.0	68.0	32.0	210.0	39.9	0.381	25.0	0.0	nan
8	3.0	173.0	82.0	48.0	465.0	38.4	2.137	25.0	1.0	nan
9	1.0	147.0	94.0	41.0	0.0	49.3	0.358	27.0	1.0	nan
10	2.0	107.0	74.0	30.0	100.0	33.6	0.404	23.0	0.0	nan

TabuLLM: AI Benchmark Report

B.3 Audit Trace: Fold 3 | Gen: 100%

#	Pregnancies	Glucose	BloodPressure	SkinThickness	Insulin	BMI	DiabetesPedigreeFunc	Age	Outcome	_fold_id
1	4.0	124.0	74.0	0.0	0.0	33.8738656464391 2	0.24344615042661 01	37.0	1.0	3.0
2	2.0	81.0	43.0	35.0	91.0	39.3716217846968 8	0.33450675844186 77	26.0	1.0	3.0
3	3.0	78.0	63.0	31.0	80.0	32.3471802022336 8	0.68751754097873 81	26.0	1.0	3.0
4	0.0	121.0	0.0	0.0	0.0	32.4909054179081	0.17027154456640 95	23.0	1.0	3.0
5	1.0	90.0	38.0	37.0	96.0	49.9872821948743 96	0.58015014957925 78	26.0	1.0	3.0
6	0.0	116.0	82.0	47.0	226.0	44.4242671667141 9	0.68367124679618 35	34.0	1.0	3.0
7	5.0	158.0	84.0	41.0	210.0	39.4	0.395	29.0	1.0	nan
8	1.0	101.0	50.0	15.0	36.0	24.2	0.526	26.0	0.0	nan
9	0.0	109.0	88.0	30.0	0.0	32.5	0.855	38.0	1.0	nan
10	5.0	147.0	75.0	0.0	0.0	29.9	0.434	28.0	0.0	nan

B.4 Audit Trace: Fold 4 | Gen: 100%

#	Pregnancies	Glucose	BloodPressure	SkinThickness	Insulin	BMI	DiabetesPedigreeFunc	Age	Outcome	_fold_id
1	11.0	140.0	82.0	42.0	324.0	39.1987424408931 06	0.52832570780868 68	57.0	1.0	4.0
2	6.0	151.0	86.0	43.0	0.0	48.7056885154581 9	0.35988721527543 44	37.0	1.0	4.0
3	1.0	93.0	64.0	31.0	160.0	37.9900170164094 5	0.67272919316995 98	23.0	1.0	4.0
4	2.0	94.0	65.0	32.0	161.0	37.7300969522194 2	0.64459477321548 39	23.0	1.0	4.0
5	7.0	144.0	110.0	0.0	0.0	46.5978279067417 06	0.23716387119965 365	39.0	1.0	4.0
6	7.0	141.0	68.0	35.0	0.0	36.0091543468764 6	0.56803911730012 89	46.0	1.0	4.0
7	5.0	117.0	86.0	30.0	105.0	39.1	0.251	42.0	0.0	nan
8	3.0	74.0	68.0	28.0	45.0	29.7	0.293	23.0	0.0	nan
9	2.0	129.0	0.0	0.0	0.0	38.5	0.304	41.0	0.0	nan
10	1.0	126.0	56.0	29.0	152.0	28.7	0.801	21.0	0.0	nan

TabuLLM: AI Benchmark Report

B.5 Audit Trace: Fold 5 | Gen: 100%

#	Pregnancies	Glucose	BloodPressure	SkinThickness	Insulin	BMI	DiabetesPedigreeFunc	Age	Outcome	_fold_id
1	0.0	141.0	61.0	29.0	132.0	37.263761266867746	0.4266339864910736	24.0	1.0	5.0
2	2.0	112.0	73.0	30.0	0.0	31.796308319360413	0.2309613392493508	24.0	1.0	5.0
3	0.0	107.0	82.0	0.0	0.0	28.773011549713214	0.6492598032504757	60.0	1.0	5.0
4	0.0	124.0	53.0	37.0	166.0	33.500130027300585	0.8101013092788525	29.0	1.0	5.0
5	3.0	125.0	66.0	24.0	186.0	31.035101787528504	0.7470390047906754	28.0	1.0	5.0
6	4.0	107.0	66.0	40.0	127.0	37.503050791845916	0.4460535405727435	24.0	1.0	5.0
7	13.0	126.0	90.0	0.0	0.0	43.4	0.583	42.0	1.0	nan
8	1.0	119.0	44.0	47.0	63.0	35.5	0.28	25.0	0.0	nan
9	1.0	181.0	78.0	42.0	293.0	40.0	1.258	22.0	1.0	nan
10	3.0	78.0	70.0	0.0	0.0	32.5	0.27	39.0	0.0	nan

B.6 Audit Trace: Fold 6 | Gen: 100%

#	Pregnancies	Glucose	BloodPressure	SkinThickness	Insulin	BMI	DiabetesPedigreeFunc	Age	Outcome	_fold_id
1	4.0	95.0	67.0	37.0	0.0	38.578104080855622	0.4362016123821753	28.0	1.0	6.0
2	2.0	90.0	68.0	41.0	0.0	38.009230449265864	0.5088310692865906	27.0	1.0	6.0
3	0.0	126.0	65.0	19.0	179.0	29.943850928259764	1.3790591522832243	26.0	1.0	6.0
4	0.0	127.0	68.0	19.0	180.0	30.563265477058714	1.3773979224323765	25.0	1.0	6.0
5	1.0	107.0	62.0	36.0	86.0	35.729030276664101	0.8286130661409509	25.0	1.0	6.0
6	7.0	153.0	74.0	27.0	127.0	31.983677457593211	0.5706418504617644	51.0	1.0	6.0
7	0.0	102.0	52.0	0.0	0.0	25.1	0.078	21.0	0.0	nan
8	4.0	76.0	62.0	0.0	0.0	34.0	0.391	25.0	0.0	nan
9	1.0	103.0	30.0	38.0	83.0	43.3	0.183	33.0	0.0	nan
10	1.0	124.0	74.0	36.0	0.0	27.8	0.1	30.0	0.0	nan

TabuLLM: AI Benchmark Report

B.7 Audit Trace: Fold 7 | Gen: 100%

#	Pregnancies	Glucose	BloodPressure	SkinThickness	Insulin	BMI	DiabetesPedigreeFunc	Age	Outcome	_fold_id
1	4.0	144.0	84.0	40.0	173.0	39.2842237797376	0.52308091341989 75	49.0	1.0	7.0
2	0.0	123.0	70.0	20.0	0.0	27.9033180965350 17	0.29570349942718 743	35.0	1.0	7.0
3	3.0	143.0	80.0	31.0	91.0	31.6695149727446 72	0.34428348797696 82	31.0	1.0	7.0
4	0.0	175.0	93.0	27.0	0.0	34.7393937229762 5	0.73002149971262 98	47.0	1.0	7.0
5	8.0	146.0	76.0	29.0	122.0	35.6156351359467 96	0.57409087459193 49	51.0	1.0	7.0
6	6.0	149.0	83.0	9.0	0.0	29.0775685698151	0.23277764192845 135	62.0	1.0	7.0
7	2.0	85.0	65.0	0.0	0.0	39.6	0.93	27.0	0.0	nan
8	2.0	142.0	82.0	18.0	64.0	24.7	0.761	21.0	0.0	nan
9	9.0	102.0	76.0	37.0	0.0	32.9	0.665	46.0	1.0	nan
10	4.0	117.0	64.0	27.0	120.0	33.2	0.23	24.0	0.0	nan

B.8 Audit Trace: Fold 8 | Gen: 100%

#	Pregnancies	Glucose	BloodPressure	SkinThickness	Insulin	BMI	DiabetesPedigreeFunc	Age	Outcome	_fold_id
1	4.0	175.0	79.0	0.0	0.0	38.4797563637720 5	0.21967122464406 733	39.0	1.0	8.0
2	9.0	111.0	70.0	27.0	0.0	27.6463562554144 53	0.15290364210704 208	38.0	1.0	8.0
3	3.0	126.0	71.0	37.0	0.0	32.1531657010907 14	0.22313922604281 552	24.0	1.0	8.0
4	7.0	110.0	67.0	0.0	0.0	29.2208986667040 73	0.86298722666935 76	33.0	1.0	8.0
5	7.0	179.0	65.0	36.0	363.0	35.6442990294476 9	0.81017961345384 66	36.0	1.0	8.0
6	1.0	153.0	78.0	51.0	96.0	47.5855881815825 9	0.61308698955431 66	28.0	1.0	8.0
7	1.0	119.0	44.0	47.0	63.0	35.5	0.28	25.0	0.0	nan
8	8.0	196.0	76.0	29.0	280.0	37.5	0.605	57.0	1.0	nan
9	1.0	118.0	58.0	36.0	94.0	33.3	0.261	23.0	0.0	nan
10	1.0	116.0	70.0	28.0	0.0	27.4	0.204	21.0	0.0	nan

TabuLLM: AI Benchmark Report

B.9 Audit Trace: Fold 9 | Gen: 100%

#	Pregnancies	Glucose	BloodPressure	SkinThickness	Insulin	BMI	DiabetesPedigreeFunc	Age	Outcome	_fold_id
1	5.0	67.0	72.0	25.0	0.0	31.068764573883897	1.12118240067797 03	33.0	1.0	9.0
2	8.0	125.0	70.0	23.0	116.0	30.549151184250118	0.44011228999506 313	42.0	1.0	9.0
3	6.0	142.0	82.0	30.0	296.0	34.032536193399814	0.47345454870810 91	58.0	1.0	9.0
4	3.0	121.0	80.0	0.0	0.0	37.56882281332021	0.61403822468785 6	24.0	1.0	9.0
5	1.0	119.0	77.0	27.0	149.0	34.765840853294335	0.53104667767010 36	35.0	1.0	9.0
6	0.0	141.0	61.0	29.0	132.0	37.218300822326655	0.42659679158190 36	24.0	1.0	9.0
7	8.0	65.0	72.0	23.0	0.0	32.0	0.6	42.0	0.0	nan
8	10.0	75.0	82.0	0.0	0.0	33.3	0.263	38.0	0.0	nan
9	2.0	146.0	70.0	38.0	360.0	28.0	0.337	29.0	1.0	nan
10	5.0	126.0	78.0	27.0	22.0	29.6	0.439	40.0	0.0	nan

B.10 Audit Trace: Fold 10 | Gen: 100%

#	Pregnancies	Glucose	BloodPressure	SkinThickness	Insulin	BMI	DiabetesPedigreeFunc	Age	Outcome	_fold_id
1	7.0	152.0	87.0	43.0	0.0	49.235114977205 68	0.34160669388728 4	36.0	1.0	10.0
2	2.0	117.0	63.0	34.0	147.0	35.332443258671 88	0.63289299993772 11	28.0	1.0	10.0
3	5.0	119.0	65.0	33.0	129.0	35.636599530027 99	0.52543803289804 06	26.0	1.0	10.0
4	4.0	134.0	82.0	35.0	85.0	32.970307320022 75	0.29737858017563	34.0	1.0	10.0
5	0.0	113.0	66.0	27.0	75.0	34.204384897166 44	0.63020768845978 47	27.0	1.0	10.0
6	0.0	113.0	68.0	30.0	91.0	35.015856402250 54	0.57640762985656 16	30.0	1.0	10.0
7	3.0	180.0	64.0	25.0	70.0	34.0	0.271	26.0	0.0	nan
8	3.0	99.0	80.0	11.0	64.0	19.3	0.284	30.0	0.0	nan
9	4.0	148.0	60.0	27.0	318.0	30.9	0.15	29.0	1.0	nan
10	7.0	136.0	90.0	0.0	0.0	29.9	0.21	50.0	0.0	nan